

PRAVIN KUMAR R

ASSOCIATE DATA SCIENTIST

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PROFESSIONAL SUMMARY

Associate Data Scientist with a solid Civil Engineering foundation (B.E., First Class) and proven hands-on experience across the full analytics lifecycle -- from raw data wrangling and exploratory analysis to baseline model implementation and stakeholder-ready interactive dashboards. Proficient in Python (Pandas, NumPy, Scikit-Learn), SQL, and Plotly/Streamlit, with a strong grounding in descriptive and inferential statistics. Adept at writing clean, documented, and reproducible pipelines that bridge raw data and actionable business strategy in FinTech and applied AI domains.

SKILLS

Programming & Data Wrangling: Python (Pandas, NumPy, Scikit-Learn), SQL, SQLite, SQLAlchemy, REST APIs, FastAPI

Statistics & EDA: Descriptive & Inferential Statistics, Hypothesis Testing, Probability Distributions, Exploratory Data Analysis, Feature Engineering

Machine Learning: Regression, Classification, Clustering, Random Forest, XGBoost, K-Means, PCA, Ridge, Elastic Net, Cross-Validation

Visualization & Reporting: Streamlit, Plotly, Matplotlib, SHAP Explainability, Automated PDF Report Generation, Git, GitHub Actions CI, Docker

PROJECTS

CryptoCast: EDA & Multi-Horizon Forecasting Pipeline

Tools: Python, Pandas, NumPy, Scikit-Learn, PyTorch, Streamlit, SHAP

- Wrangled and validated **~5,000 daily Bitcoin records (2010–2024)**, engineering 60-day rolling window features and running EDA to surface volatility clusters, trend anomalies, and on-chain distribution patterns.
- Implemented and evaluated 5 forecasting architectures under chronological Walk-Forward cross-validation, selecting the optimal model achieving **2.06% MAPE at 1-day** and 5.93% at 7-day horizons.
- Deployed an **11-page interactive Streamlit analytics studio** with SHAP feature importance, translating model outputs into plain-language, stakeholder-ready business intelligence reports.

Nifty 50 Stock Analysis & Sector Intelligence Dashboard

Tools: Python, SQL, SQLAlchemy, Pandas, Plotly, Scikit-Learn

- Queried and aggregated Nifty 50 historical OHLCV data using **SQLAlchemy and multi-table SQL joins**, performing null-value audits, outlier detection, and consistency checks to ensure data integrity across 50 stocks.
- Conducted sector-level **exploratory data analysis** -- computing rolling returns, volatility metrics, and correlation matrices -- to identify cyclical vs. defensive sector behavior and flag performance anomalies.
- Designed interactive **Plotly dashboards** with filterable trend charts and correlation heatmaps, documenting all query logic and pipeline steps in reproducible, version-controlled notebooks for team collaboration.

Other Verified Projects: ERIA Multi-Agent RAG System (Groq LLaMA 3.3 70B, DistilRoBERTa), HuBMAP (Swin-TransUNet, 0.6782 Dice), Soil Grain Size Estimation (55.22 OOF Weighted EMD) -- available on GitHub.

PROFESSIONAL EXPERIENCE

Data Science Intern — Internship Studio

Jan 2026 – Feb 2026 | Remote

- Collected, preprocessed, and validated **14,102 Fitbit telemetry records**, performing categorical encoding, StandardScaler normalization, and 80/20 train-test splitting to ensure data integrity and consistency.
- Built, tested, and evaluated baseline **regression and clustering models** (Linear Regression, Random Forest, K-Means) -- reducing caloric prediction MAE from 34.97 to **3.78 kcal (89.2% error reduction, $R^2 = 0.9979$)**.
- Documented the full ML pipeline in reproducible notebooks, applying **PCA + K-Means** (Silhouette 0.4546) to segment users into 4 exertion zones and containerizing inference via Docker and FastAPI REST endpoints.

EDUCATION

Master Data Science Program — HCL GUVI, IIT Madras Research Park Incubated

Completed Jun 2026

B.E. Civil Engineering (First Class Honors) — Rajalakshmi Engineering College, Anna University

Aug 2017 – May 2021

CERTIFICATIONS

Certifications: Data Science & Analytics (HCL GUVI), Build & Deploy AI Apps with Google AI Studio, ML for Earth Observation (EO College), Machine Learning & Pandas (HCL GUVI)